Revitalizing U.S. Transportation: Reducing Auto Dependence and Embracing Transit-Oriented Design

Presented to
US Department of Transportation

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EXECUTIVE SUMMARY[1]

POLICY QUESTION (Page 1)

What infrastructure policies should the U.S. adopt to catch up to other developed nations in terms of the availability and reliability of regional and national public transportation, such as high-speed rail?

RECOMMENDATION (Page 13)

The report recommends that the Department of Transportation prioritize the development of urban subway and light rail where appropriate, and promote transit-oriented design in city planning

PROBLEM STATEMENT (Page 1)

The United States has an outdated transportation system that is overly reliant on automobiles, suffers from inefficient budget allocation, lacks effective public transportation, and is characterized by insufficient interconnectedness.

Historically, the US government allocates the majority of the transportation budget to the construction and the maintenance of highways. Since the 1920s, Americans have become more reliant on cars. In 2022, around 92 percent of American households have at least one working car.[2] After the construction of the interstate highways, the budget for the Federal Highway Administration became significantly higher than the Federal Railway Administration and Federal Transit Administration.

To address regional rail the US can look towards China, Japan, and Switzerland as examples for how to develop and manage a national rail system. China has a rapidly expanding rail network that takes advantage of economies of scale for construction efficiency. Japan has centralized their rail management system and used a vertical integration and regional districts for optimal management. Switzerland has focused on creating a timetable for easy transfers to cut down on trip length and has also implemented a unique funding structure.

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[1] *This student paper was prepared in 2024 in partial completion of the requirements for Public Policy 803, a course in the Masters of Public Policy Program at the Sanford School of Public Policy at Duke University. The research, analysis, and policy alternatives and recommendations contained in this paper are the work of the student team that authored the document, and do not represent the official or unofficial views of the Sanford School of Public Policy or of Duke University. Without the specific permission of its authors, this paper may not be used or cited for any purpose other than to inform the client organization about the subject matter. The authors relied in many instances on data provided to them by the client and related organizations and make no independent representations as to the accuracy of the data.*

[2] Ashlee Valentine and Kelly Anne Smith, “Car Ownership Statistics 2024,” Forbes, July 15, 2024, https://www.forbes.com/advisor/car-insurance/car-ownership-statistics/.

In the US, cities often lack interconnectedness, they are designed for transportation via car and can make travel unsafe for pedestrians and cyclists. Companies are moving towards the suburbs which often do not have public transit routes. Suburbs are also designed in a spiral grid system that is unfriendly for pedestrians and cyclists. This makes them inaccessible to people without a reliable car.

In the U.S. a common problem is that where there is public transport, ridership is low. This is because routes are poorly designed and often exclude those who use public transport for trips that are not oriented to employment. This design excludes older people, women and young students.

Low ridership creates a funding problem, as without users there is no revenue and the system continues to deteriorate without this investment. There are federal programs available that prioritize public transport with an emphasis on energy-efficient systems.

CRITERIA (Page 7)

1. Minimize reliance on highways.
2. Maximize the use of public transit, local and interstate.
3. Establish rail competition with other transportation methods.
4. Secure long-term sustainability.
5. Maximize interconnectedness within urban regions.
6. Ensure robust safety measures for travelers and pedestrians.

ALTERNATIVES (Page 7)

- Expand Amtrak Routes
- Establish Urban Subway lines, Light Rail or Bus Systems
- Incorporate the concept of Transit Oriented Designs in City Planning
- Adopt Pedestrian-Bicyclist-Friendly Designs

POLICY QUESTION

What infrastructure policies should the U.S. adopt to catch up to other developed nations in terms of the availability and reliability of regional and national public transportation, such as high-speed rail?

PROBLEM STATEMENT

The History of Auto-Dependency

The United States has a central problem of auto-reliance. In the 1920s, Henry Ford revolutionized transportation by making cars affordable for average Americans with the Model T. This shift was further solidified by the Federal-Aid Highway Act of 1956, signed by President Eisenhower, which led to the construction of interstate highways, an infrastructure project deeply ingrained in American car culture. Rail infrastructure suffers from a lack of federal investment; for instance, while the Inflation Reduction Act allocated $50 billion to support cars, only a fraction went to alternative transit. Additionally, the federal government funds just 25% of infrastructure projects, far less than many European countries, leaving states with limited resources and facing high land and construction costs.

Since then, roads have been designed primarily to accommodate cars rather than people. As of 2022, 91.7 percent of American households owned at least one working car,[3] with 80% of Americans commuting by driving and only 3% using mass transit.[4] However, with an aging population and a younger generation increasingly seeking sustainable alternatives, the demand is growing for an interconnected transportation system that is efficient, reliable, and affordable. Moreover, riding a train instead of driving a car for medium-length distances can reduce emissions by approximately 80%.[5]

Highway Spending Furthering American Auto-Reliance

Due to the history of auto-dependency, the US government continues to prioritize investment in highways rather than mass transit and rail systems. As illustrated by these two graphs from the *New York Times* (Figure 1),[6] The last major

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[3] Ashlee Valentine and Kelly Anne Smith, “Car Ownership Statistics 2024,” Forbes, July 15, 2024, https://www.forbes.com/advisor/car-insurance/car-ownership-statistics/.
[4] Brad Plumer and Nadja Popovich, “America Has Long Favored Cars over Trains and Buses. Can Biden Change That?,” The New York Times, April 2, 2021, https://www.nytimes.com/2021/04/02/climate/biden-public-transit-amtrak.html.
[5] Hannah Ritchie and Max Roser, “Which Form of Transport Has the Smallest Carbon Footprint?,” Our World in Data, March 18, 2024, https://ourworldindata.org/travel-carbon-footprint.
[6] Brad Plumer and Nadja Popovich, “America Has Long Favored Cars over Trains and Buses. Can Biden Change That?,” The New York Times, April 2, 2021, https://www.nytimes.com/2021/04/02/climate/biden-public-transit-amtrak.html.

federal investment from the federal government in mass transit and rails was back in the 1980s, even then, spending on highways was comparable or greater. State and local governments increased investments in mass transit significantly since the 2000s, but these investments remain far below those allocated to highways. In recent years, there has been a shift of interest with more funds allocated to railway and transit. However, the numbers remain significantly lower compared to highways (Figure 2). For fiscal year 2024, around $68.8 million was allocated to the Federal Highway Administration (FHWA), while the combined funding for the Federal Railway Administration (FRA) and Federal Transit Administration (FTA) totaled only $40.6 million, roughly 59% of FHWA's budget.[7]

The Everyday Consumer

In 2018, 31.7 million people traveled using Amtrak services marking a 10% increase since 2008. While 777.9 million people took domestic flights. On websites that compare flights and train tickets for the same routes, results found that the top 100 Amtrak routes booked train travel was 80 percent cheaper, with travelers saving an average of $132.[8] Additionally, the shorter overall travel time represents an advantage in the option for the train. It is worth noting that these benefits stop when trips are longer than 300 miles, according to the New York Times.[9]

What Can We Learn from European and Asian Countries?

Overall, looking at the EU system and China and Japan, in order for consumers to choose rail transit, trains must be fast and reliable, it needs to be easy to buy tickets, and it must be easy to access stations. Faster and more reliable trains will come out of infrastructure improvements. Reliability also comes with managerial improvements. Easy, centralized, online ticketing comes as a result of strategically centralizing railway management. Finally, stations need to be able to handle high capacity and also need to be connected to other resources, such as urban developments and regional transit, which we discuss in a later section.

China Has Efficient Infrastructure Design

China has, by far, the largest and fastest developing high-speed rail network. How has China developed its high-speed rail at a rate no other country has? Since 2008, China has developed over 50,000km of high-speed rails. By comparison, the top 5 countries after China have a combined 11,000km of high-speed rail.[10] While part of China's rapid rail development is because the cost of labor there as well as

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[7] "Fiscal Year 2024 Budget Estimates," U.S. Department of Transportation, accessed December 11, 2024, https://www.transportation.gov/mission/budget/fiscal-year-2024-budget-estimates.
[8] https://www.concur.com/en-us/concur-hipmunk-faq
[9] Elaine Glusac, "Train vs. Plane: And the Winner Is ... Well, It Depends," The New York Times, September 20, 2019, https://www.nytimes.com/2019/09/20/travel/trains-airplanes.html.
[10] Wong, Dennis. "How China Built the World's Largest Bullet Train Network." South China Morning Post, November 24, 2022. https://multimedia.scmp.com/infographics/news/china/article/3200811/high-speed-railway/index.html.

land rights and relocation are cheaper, a huge part of that is rail development technology in China.

China uses standardized design plans to keep the cost of physical infrastructure development low in rail expansion. A prime example is viaducts; Chinese railways heavily rely on viaducts to elevate rail lines and minimally impact the agricultural land below. Viaduct design is standardized, down to the dimensions of each arch and pillar.[11] Then viaduct parts are mass-manufactured and can be sent to any project, taking advantage of economies of scale to reduce production costs. In addition, by using one standardized design, the cost of designing and testing is amortized over all projects that use that part, further reducing the per-unit cost.

China's commitment to long-term rail development also helps it take advantage of economies of scale. By investing in multiple projects at once, expensive specialized machines can be shared across projects, bringing down total costs.[11] In 2008, China released a 12 year rail development plan.[11] This commitment to rail development allowed contractors the security to invest in technologies they knew would be used for the next 12 years. If the government was unable to set long term goals, contractors would be hesitant to invest in innovative technology given that the government could switch priorities to another sector within a few years. For example, China initially imported the slab track manufacturing process from Germany. Because contractors were able to invest in manufacturing technology, they were able to lower the cost of the slab track manufacturing process to ⅓ of the original cost.[11] If the US were able to commit to large-scale rail development, it too could take advantage of such efficiencies.

Operational Efficiency with Vertical Management and Nationalization

Organizationally, other countries have seen the best results with top-down government-owned rail systems with vertical integration and regional divisions. Based on the Japanese model, regional division is where the country is split into geographic regions that manage their own internal train systems.[12] Vertical integration refers to where each region's rail management owns the rolling stock and tracks, employs the workers, and operates the trains.[12] By comparison, Amtrak currently owns and operates trains, but over 70% of the railway tracks they operate on are owned by a different company.[13] Amtrak pays to use the tracks but other companies, including freight companies, also use the tracks, which can lead to scheduling conflicts and delays.

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[11] Ollivier, Gerald. "How China Builds High-Speed Rail for Less." International Railway Journal, October 9, 2014. https://www.railjournal.com/in_depth/how-china-builds-high-speed-rail-for-less/.
[12] Eoin Quill, Andrzej Ancygier, Olivia Waterton, Sylvia Schmidt, Mallika Singh. "Railways development: lessons for the EU." July 2023. Climate Analytics; Berlin. https://ca1-clm.edcdn.com/assets/4i-TRACTION_Spin_off_2023_D3.4_Railway-development.pdf?v=1697189676.
[13] Amtrak. "FY 2023 Company Profile." Amtrak, 2023. https://www.amtrak.com/content/dam/projects/dotcom/english/public/documents/corporate/nationalfactsheets/Amtrak-Company-Profile-FY2023-041824.pdf.

In the Japanese system of vertical integration and regional management, each region has full control over their scheduling, to allow for efficient resource management. Additionally, conductors only operate trains on stretches of tracks that they are familiar with, increasing safety and speed.[14] This is an important model for the US which is geographically large and where railway lines pass through many different states. The geography of the US lends itself to establishing each state as its own rail region which coordinate to manage trans-continental lines. While Japan is a prime example of the benefits of vertical integration, Austria also has a successful vertically integrated rail system and the UK's rail system declined when it moved away from vertical organization.

In addition to vertical management, rail systems need to be government owned in order to maximize operational efficiency. Privatization of rail in Europe has largely failed to increase spending efficiency and ridership because companies are focused on profit.[15] When companies treat workers as costs, their reputation as a good employee fails and they have difficulties recruiting and retaining employees.[16] Employees are a necessary resource to ensure the smooth operation and maintenance of rail systems. In order for rails to remain cost effective and run smoothly, rail needs to be seen as an attractive employer.

Clock Face Timetables for Easy Transfers

One lesson from Swiss rail reorganization is the benefits of timetable improvements. The Swiss designed a Clock Face Timetable, where all trains arrive at interchange stations just before the hour mark, and all trains leave those stations just after the hour mark.[14] That means all trains, regardless of final destination, are available for passengers to transfer to when they arrive at an interchange station. This allows riders to easily catch the trains they need, increasing connection efficiency and shortening overall trip time. By using a Clock Face Timetable, Switzerland has made rail competitive with other forms of domestic transportation.

Use Road Taxes to Fund Rail Expansion

Where does the money come to fund rail improvements and expansion? In Switzerland, there is a heavy tax on freight trucks, which goes directly to funding the expansion of freight rail.[14] Comparatively, in other countries including the US, the gains from road taxes are often put back into improving road infrastructure, instead of going to rail.[14] In other countries and the EU, rail development is set as a priority

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[14] Eoin Quill, Andrzej Ancygier, Olivia Waterton, Sylvia Schmidt, Mallika Singh. "Railways development: lessons for the EU." July 2023. Climate Analytics; Berlin. https://ca1-clm.edcdn.com/assets/4i-TRACTION_Spin_off_2023_D3.4_Railway-development.pdf?v=1697189676.

[15] "Policy Position Paper Lessons Learned from Three Decades of Liberalisation of the European Railways." European Transport Workers' Federation, 2021.

[16] "Shifting Mobility to Rail. Lessons from Rail Liberalisation and a New Direction for People and Planet." AKEUROPA, April 26, 2024. https://www.akeuropa.eu/en/shifting-mobility-rail-lessons-rail-liberalisation-and-new-direction-people-and-planet.

so it is easier to find funding for it.[17] A combination of prioritizing rail development and reallocation road taxes can help fund rail development in the US.

Lack of Interconnectedness

Cities in the US often lack "interconnectedness." Road designs tend to accommodate cars over people, making travel unsafe for pedestrians and cyclists. This leads to a dependency on cars, which becomes increasingly unsustainable. Studies show that public transportation, including buses and subways, is safer than driving a car. With a portion of the American population aging, many no longer can drive safely. Meanwhile, the younger generation, particularly college-educated individuals and urban residents, is showing a preference for eco-friendly transportation options that are interconnected to the city design[18].

In recent years companies have tended to move towards suburbs.[19] Suburbs are designed for cars, they are connected to other areas of the town or city via highway. The placement of roads is curve-like to reduce the speed of cars but it also makes them less pedestrian friendly as walking along a spiral is longer than walking a straight line.[20] Highways, spiraling roads, and cul-de-sacs make it difficult to reach bus stops, metro stations, and other destinations on foot- as is visible in Figure 3[21]. It also means that certain companies are only an employment option for people who can access them via car since often there is no public transport running through them.

When there is public transport there is often low ridership, this is often attributed to a lack of interest or usefulness but studies have shown that the lack of ridership is due to it not responding to citizen's needs.[22] But when ridership decreases so does investment and revenue, which diminishes the quality of service and creates a cycle of low ridership and lack of funding.[23] When the routes don't make sense for people they will choose to not use them, this is often due to poor

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[17] Otsuka, Noriko, Felix Christian Günther, Ilaria Tosoni, and Cecilia Braun. "Developing Trans-European Railway Corridors: Lessons from the Rhine-Alpine Corridor." *Case Studies on Transport Policy* 5, no. 4 (December 2017): 527–36. https://doi.org/10.1016/j.cstp.2017.06.005.
[18] Andy Hong, "The City Looks Different When You're Older," The New York Times, September 6, 2023, https://www.nytimes.com/interactive/2023/09/06/opinion/seniors-urban-planning.html.
[19] Chelsea Graham, "Why Doesn't the U.S. Have Better Public Transportation?," Sustainable America, April 30, 2023, https://sustainableamerica.org/blog/why-doesnt-the-us-have-better-public-transportation/.
[20] Joseph Stromberg, "The Real Reason American Public Transportation Is Such a Disaster," Vox, August 10, 2015, https://www.vox.com/2015/8/10/9118199/public-transportation-subway-buses.
[21] Joseph Stromberg, "The Real Reason American Public Transportation Is Such a Disaster," Vox, August 10, 2015, https://www.vox.com/2015/8/10/9118199/public-transportation-subway-buses.
[22] Pietro Nivola, "Are Europe's Cities Better?," Brookings, July 28, 2016, https://www.brookings.edu/articles/are-europes-cities-better/.
[23] Jonathan English, "Why Is American Mass Transit so Bad? It's a Long Story.," Bloomberg.com, August 31, 2018, https://www.bloomberg.com/news/features/2018-08-31/why-is-american-mass-transit-so-bad-it-s-a-long-story.

design[24]. The Brookings Institute analyzed Chicago's routes and a route that would take 20 minutes by car takes 60 minutes via public transit.[25] As seen in Figure 3, globally cities with the highest ridership operate at a 15 minute frequency while most American cities operate at a 30 minute frequency and report less ridership.[26]

Additionally, routes are often planned for people, often men, who have a direct, point-to-point daily work journey. The system is designed around mobility related to employment. If a household owns a car, it is the men who dominate access to it.[27] While women are more likely to walk or use public transport, in US cities Philadelphia and Chicago women represent 64% and 62% of users respectively.[28] Women need to accomplish multiple objectives within the same overarching journey, dropping kids off, household management tasks as well as caregiving tasks. Criado Perez refers to this as a trip-chain, tying smaller trips into a larger journey plan. And the most efficient answer to this need is bus routes, as they can plan for more stops than intercity rail.

When the federal government provides funding it is limited to big city projects. Operations that run local bus and train routes that could be a viable alternative for people are constantly underfunded. The idea that public transportation can only be a priority in dense cities is false. States and local governments can afford the annual expenditure to run buses reasonably often, every day and for more people, not just those using it for employment commuting.[29] Another complication for funding is that in order to further reduce greenhouse emissions, policymakers must ensure that both rail and bus options become energy-efficient, which may increase the initial investment cost. Acts and programs have prioritize this when awarding funding, like the Federal-State Partnership for Intercity Passenger Rail (FSP)[30] Grant Program and the Bipartisan Infrastructure Law (BIL).

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[24] Joseph Stromberg, "The Real Reason American Public Transportation Is Such a Disaster," Vox, August 10, 2015, https://www.vox.com/2015/8/10/9118199/public-transportation-subway-buses.
[25] Chelsea Graham, "Why Doesn't the U.S. Have Better Public Transportation?," Sustainable America, April 30, 2023, https://sustainableamerica.org/blog/why-doesnt-the-us-have-better-public-transportation/.
[26] Jonathan English, "Why Is American Mass Transit so Bad? It's a Long Story.," Bloomberg.com, August 31, 2018, https://www.bloomberg.com/news/features/2018-08-31/why-is-american-mass-transit-so-bad-it-s-a-long-story.
[27] Caroline Criado-Perez, *INVISIBLE WOMEN: Data Bias in a World Designed for Men* (Ashland, OR: Blackstone Publishing, 2019).
[28] Nicole Badstuber, "Mind the Gender Gap: The Hidden Data Gap in Transport," London Reconnections, December 5, 2019, https://www.londonreconnections.com/2019/mind-the-gender-gap-the-hidden-data-gap-in-transport/.
[29] Jonathan English, "Why Is American Mass Transit so Bad? It's a Long Story.," Bloomberg.com, August 31, 2018,
[30] Federal-State Partnership for Intercity Passenger Rail (FSP) Grant Program," Federal-State Partnership for Intercity Passenger Rail (FSP) Grant Program | FRA, November 15, 2024, https://railroads.dot.gov/federal-state-partnership-intercity-passenger.

CRITERIA

1. **Minimize reliance on highways.** Reduce auto-dependency by encouraging the use of public transportation. Shifting focus away from highways will allow greater funding for the FRA and the FTA.

2. **Maximize the use of public transit, local and interstate.** Promote interconnected public transit at both local and interstate levels to create an efficient, seamless network.

3. **Establish rail competition with other transportation methods**. Ensure rail systems are well connected, fast, reliable, and affordable to compete effectively with other transportation methods.

4. **Secure long-term sustainability.** Secure policies that are financially viable and environmentally sustainable to ensure lasting impact.

5. **Maximize interconnectedness within urban regions**. Subway and bus stations must respond to diverse commuting needs and be established near residential, workplace, and commercial areas.

6. **Ensure robust safety measures for travelers and pedestrians.** Infrastructure must protect pedestrians and cyclists to ensure safety for all travelers, and mass transit must be at least as safe as driving.

ALTERNATIVES

- **Expand Amtrak Routes:** Despite its current inefficiencies, Amtrak remains the most widely used rail system in the US. Integrating the lessons we learn from other countries can enhance the passenger experience and encourage broader use.
    - **Nationalize rail management** under the federal government and use vertical integration and regional separation for more efficient management.
    - **Invest in rail infrastructure expansion** through long term investment, standardized design, and shared machines, with a focus on safety, noise reduction, and maintainability.
    - **Invest in key supporting technologies** including a unified ticketing system and a clock face time table.
    - **Fund** the expansion through carbon taxes on air or car travel.

- **Establish Urban Subway lines, Light Rail or Bus Systems:** Adding subway lines, light rail or bus systems in urban regions enhances interconnectedness. For example, key areas like the Research Triangle Park (RTP) and North Carolina's economic center that is

Charlotte would benefit from a more integrated transportation network. This can be done through investment in transportation infrastructure that answers communities needs, light trains aren't the only option, buses offer a more affordable option for local governments. Offering transportation options that respond to the communities' needs can look different to each city or town

- **Incorporate the concept of Transit Oriented Designs (TOD) in City Planning:** TOD is defined as "a mixed-use community that encourages people to live near transit services and to decrease their dependence on driving."[31] This alternative promotes the use of public transportation by strategically placing retail and services near transit hubs.

- **Adopt Pedestrian-Bicyclist-Friendly (PBF) Designs:** The PBF concept, by Dr. Susan Handy, focuses on designing cities where walking and cycling are safe, comfortable, and feasible. Guided by three principles of distance, protection, and integration, this alternative aims to ensure short commutes are walkable, provide robust protection for pedestrians and cyclists, and integrate walkways and cycling paths into urban planning and transportation networks.[32]

ANALYSIS

We will be scoring each alternative based on how well it performs on each equally weighted criterion using the following rubric.

Score	Explanation
0	Fails this criterion
1	Partially satisfies criterion
2	Satisfies criterion

[31] Ian Carlton, "Histories of Transit-Oriented Development: Perspectives on the Development of the Tod Concept," eScholarship, University of California, September 2, 2009, https://escholarship.org/uc/item/7wm9t8r6.
[32] Susan Handy, "Making US Cities Pedestrian- and Bicycle-Friendly," Transportation, Land Use, and Environmental Planning, 2020, 169–87, https://doi.org/10.1016/b978-0-12-815167-9.00009-8.

Alternative 1: Expand Amtrak Routes

Alternative	Minimize highway use	Maximize public transit use	Make rail competitive	Sustainable long term	Maximize urban interconnectedness	Safety	Total Score
Expand Amtrak Routes	**2**	**2**	**2**	**2**	**0**	**1**	**9**

Expanding Amtrak routes allows more people to use Amtrak as an alternative to driving for interstate travel. This alternative satisfies criterion 1.

Part of this alternative involves nationalizing Amtrak, which would make Amtrak public transportation. Thus as people use Amtrak more, we maximize the use of public transit in interstate travel. If the local public infrastructure at Amtrak stations is good, then increasing Amtrak ridership also increases local public transit usage. This alternative satisfies criterion 2.

Amtrak expansion as described for this alternative will help make rail competitive with other modes of transportation. Investing in physical infrastructure and building out the Amtrak network will help riders access destinations they need to get to. Changing the management structure and implementing a Clock Face Timetable will allow rail transportation to run more smoothly and efficiently, based on what we've seen in Switzerland. This will minimize delays from sharing tracks and make it easier for riders to schedule transfers, helping make Amtrak faster and more reliable. Creating a unified ticketing system means it will be easy for riders to compare and purchase tickets online. And as seen in multiple European countries, having the government run Amtrak and create a good working environment for employees helps cut long term costs and maintain smooth operations, bringing costs down for consumers and also contributing to reliability. This alternative fully satisfies criterion 3.

Expanding Amtrak using the financial model that the Swiss use will help secure long term financial sustainability. The US can take revenue from carbon taxes that target rails' competitors including driving and flying, and use that money to invest in rail. This does not increase taxes for anyone; it simply reallocates what taxes are ready collected to invest in cleaner transportation methods instead of investing in carbon heavy and environmentally detrimental transportation methods. In this way, expanding Amtrak also supports long term environmental sustainability. This alternative satisfies criterion 4.

Because Amtrak is a regional rail meant to connect different urban areas, it will not affect the interconnectedness within urban regions, so it fails to satisfy criterion 5.

Traveling by train is a safe method of transportation, and expanding Amtrak also requires investing in safe rail travel, so this alternative ensures robust safety measures for travelers. However, as discussed above, Amtrak is primarily a regional rail system and will have little to no impact on pedestrian safety. Thus this alternative only partially satisfies criterion 6.

Alternative 2: Establish Urban Subway lines, Light Rail or Bus Systems

Alternative	Minimize highway use	Maximize public transit use	Make rail competitive	Sustainable long term	Maximize urban interconnectedness	Safety	Total Score
Add subway, light rail or bus system	2	2	1	1	2	2	10

Establishing public transport systems minimizes the reliance on highways as creating a system that aligns with citizen's needs increases ridership even when cars remain an option. In order for it to be an attractive option the service must run frequently throughout the entire week with routes that respond to all of the population's needs, as opposed to those who only use it to commute to work. The most efficient systems run on a 15 minute frequency as opposed to how most American cities run, on a 30 minute frequency.

It maximizes the use of public transit, and it can increase the use of interstate systems and provide a more efficient, affordable and seamless network for people if the routes are planned to connect buses and intercity rails to local travel hubs- bus stations, Amtrak stations and airports.

If this condition is met then it also establishes rail as a competitive method of transportation compared to other methods. As explored previously Amtrak represents a more efficient and affordable option to flying as it implies savings in airport transit time, security lines and deposits people in city centers versus in city limits.

In order for this option to be environmentally sustainable state and local agencies must create energy efficient systems, this means energy-efficient vehicles and schedules that synchronize to optimize energy use, viaduct design, economies of scale, and keeping in mind a possible future of vertical integration. For financial sustainability, there are initial investment opportunities like the FSP and BIL. For longer-term financial security, agencies or policymakers must increase ridership, make employment opportunities attractive, and restructure tax revenue from freight carriers and roads.

Designing routes that reach suburbs tie smaller trips into a larger journey plan, and maximize the interconnectedness within urban regions as it accommodates diverse needs.

Transportation via buses and subways is significantly safer than driving, even so safety can be enhanced in the system. The system can adapt the Japanese model where conductors, either in subways or buses, only drive routes they are familiar with.

Alternative 3: Incorporate the concept of Transit Oriented Designs (TOD) in City Planning

Alternative	Minimize highway use	Maximize public transit use	Make rail competitive	Sustainable long term	Maximize urban interconnectedness	Safety	Total Score
Use transit oriented design	**2**	**2**	**2**	**1**	**2**	**2**	**11**

TOD satisfies all criterions. By centering work, life, and shopping around public transit systems, such as light rail, buses, and subway lines, TOD minimize the use of highways for daily commutes. This shift encourages reliance on mass transit for everyday destinations, promoting reduced car dependency.

TOD also maximizes public transit usage by incorporating transit stations into mixed-use developments. These developments will be designed to ensure convenience and accessibility, making destinations easily reachable.

TOD also enhances rail competitiveness by connecting light rail stations to local communities, creating seamless links between transit systems and residential or commercial areas.

Additionally, TOD promotes long-term sustainability by fostering self-sufficient communities that integrate essential daily needs within transit-accessible areas. However, its application may be limited for rural or sparsely populated areas where mass transit infrastructure is less practical.

Another benefit of TOD is the potential reduction in car-related accidents, improving safety for both regional and interstate travel. By using mass transit for daily commute, TOD also helps decrease car-related accidents and alleviate traffic congestions in the urban areas.

Alternative 4: Adopt Pedestrian-Bicyclist-Friendly (PBF) Designs

Alternative	Minimize highway use	Maximize public transit use	Make rail competitive	Sustainable long term	Maximize urban interconnectedness	Safety	Total Score
Pedestrian & cyclist design	**1**	**2**	**1**	**1**	**2**	**2**	**9**

PBF will have a minimal impact on reducing highway reliance and rail competitiveness, as they primarily focus on local infrastructure for pedestrians and cyclists.

PBF maximizes public transit usage by enabling safe and convenient access to transit stations for non-drivers who need to travel longer distances. Features such as walkable roads, streetlights, and clear signage will enhance urban interconnectedness and pedestrian safety.

By facilitating easier access to public transit, PBF can contribute to a more efficient transportation system with fewer traffic jams. It also strengthens urban interconnectedness allowing pedestrians and cyclists to reach their destinations or access public transits without relying on cars.

In conclusion, while PBF satisfied criterion 2, 5, and 6, enhancing safety, urban connectivity, and pedestrian mobility, it only partially satisfied 1, 3, and 4. This is because PBF emphasizes local travel over long-distance or regional systems on broader transit networks like interstate travel. Despite its localized focus, PBF offers significant potential for improving community: level mobility and fostering safer, more walkable communities, especially when combined with TOD.

RECOMMENDATION

Alternative	Minimize highway use	Maximize public transit use	Make rail competitive	Sustainable long term	Maximize urban interconnectedness	Safety	Total Score
Expand Amtrak Routes	2	2	2	2	0	1	9
Add subway, light rail or bus system	2	2	1	1	2	2	10
Use transit oriented design	2	2	2	1	2	2	11
Pedestrian & cyclist design	1	2	1	1	2	2	9

We recommend the DOT focus on adding urban subway and light rail as appropriate, and employing transit oriented design in city planning.

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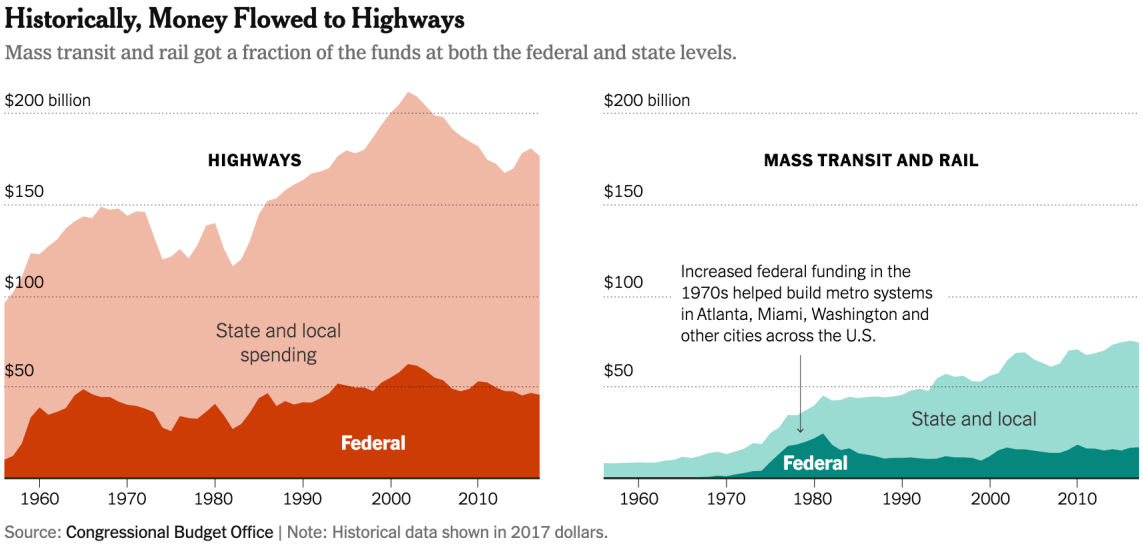
APPENDICES

Appendix 1: Historical Budget Allocation Trend

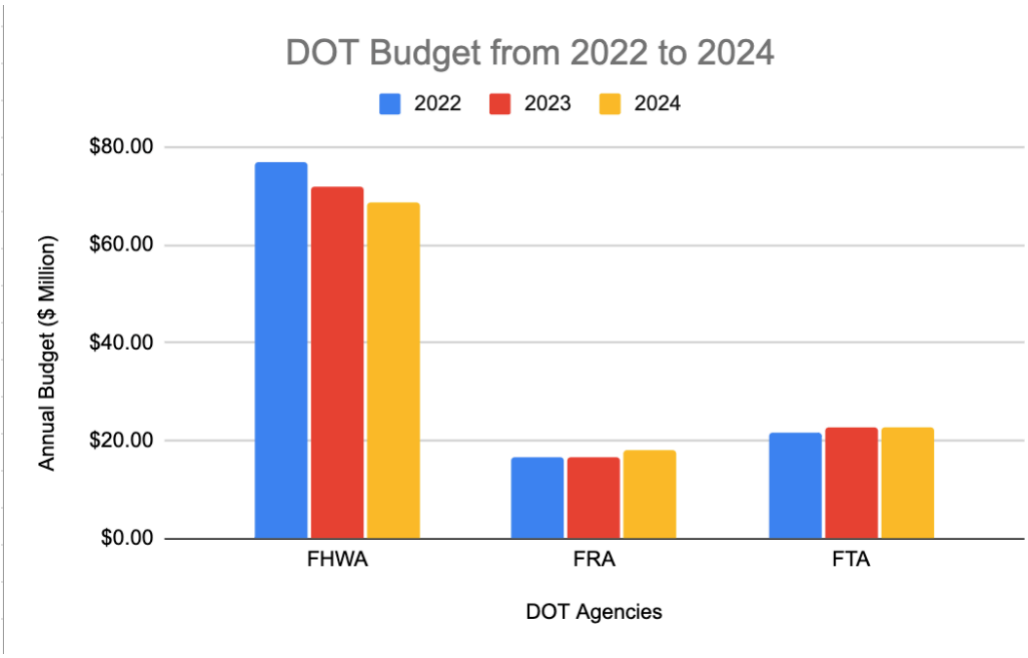
Figure 1: Historical Budget Allocation Trend for FHWA vs. FRA and FTA[33]



[33] Brad Plumer and Nadja Popovich, "America Has Long Favored Cars over Trains and Buses. Can Biden Change That?," The New York Times, April 2, 2021, https://www.nytimes.com/2021/04/02/climate/biden-public-transit-amtrak.html.

Appendix 2: Shift of Interest in Budget Allocation

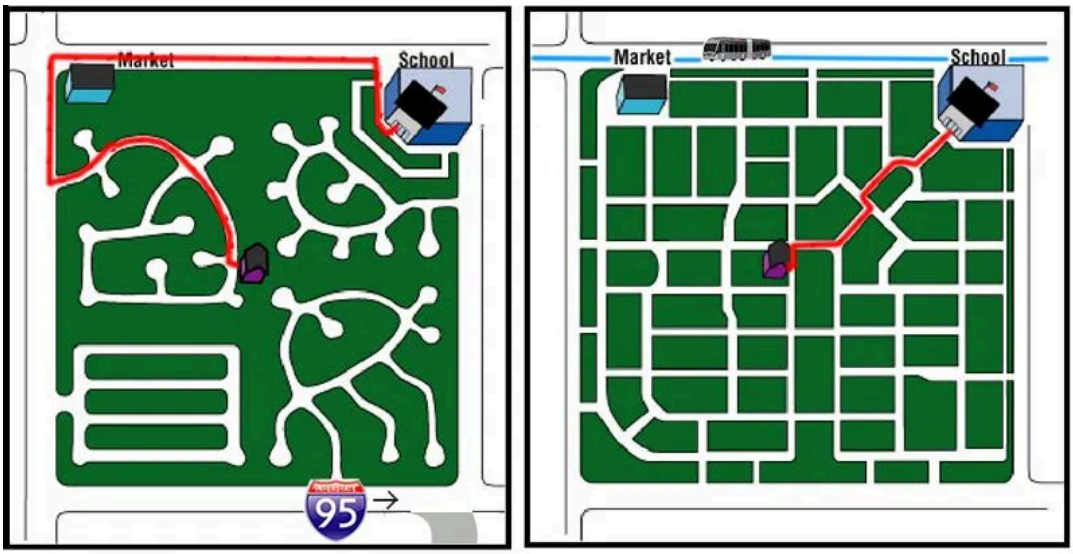
Figure 2: Shift of Interest in Budget Allocation (2022 - 2024)[34]



[34] Designed by Team, with data from Fiscal Year 2024 Budget Estimates of US DOT

Appendix 3: Different Grid Design Systems

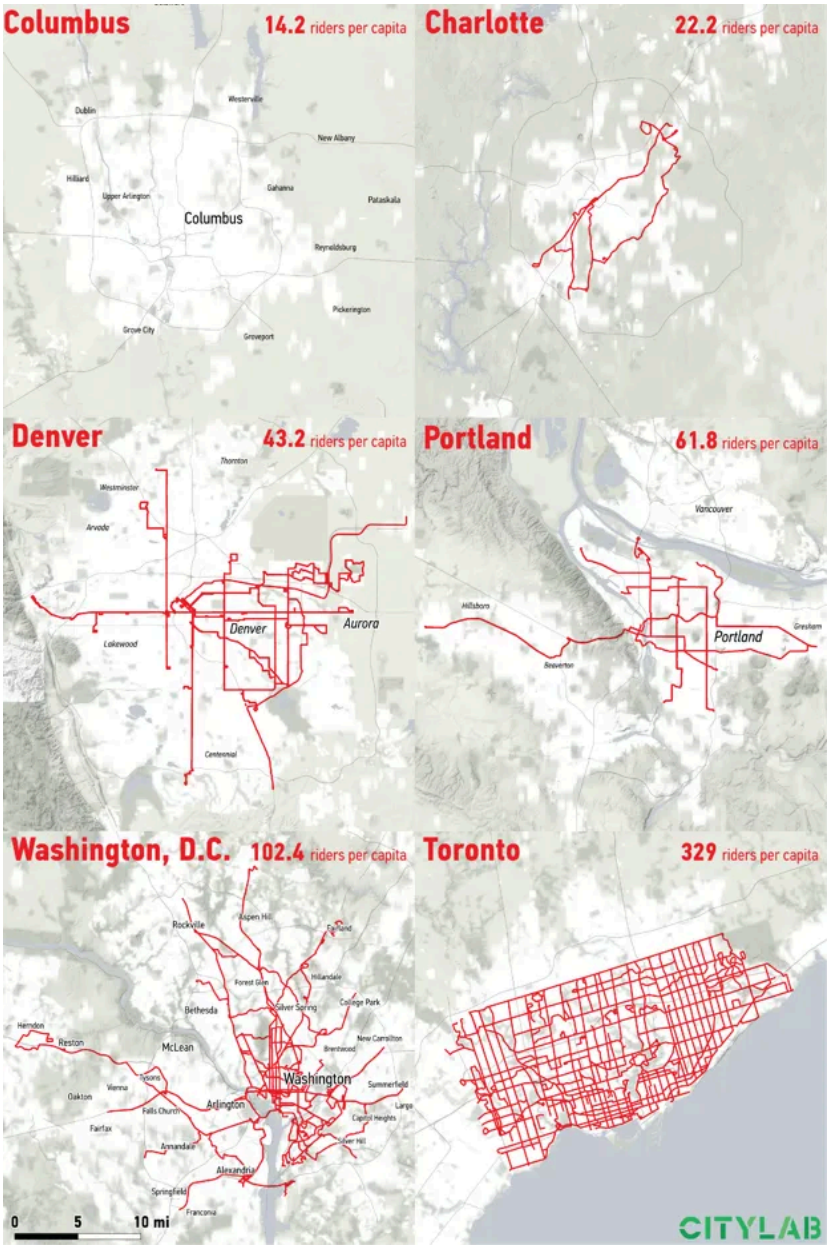
Figure 3: Different Grid Design Systems[35]



Driving-only transportation pattern Walkable connected transporation network

[35] Joseph Stromberg, "The Real Reason American Public Transportation Is Such a Disaster," Vox, August 10, 2015, https://www.vox.com/2015/8/10/9118199/public-transportation-subway-buses.

Figure 4: Different Grid Design Systems[36]



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